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CAPSTONE PROJECT

Analysis of Factors Affecting Student Productivity

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
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- References

PROBLEM STATEMENT:

- Students today face many distractions specially from smartphone. They use social media, paly video games and many other activities. At the same time, factors like study habits, stress level, sleep hours and other life choices can significantly influence academic productivity greatly.
- The challenge is to understand how these different factors affect student performance. Without proper analysis, it is difficult to identify which behaviors improve productivity and which ones reduce focus and academic efficiency.

PROPOSED SOLUTION:

- The proposed solution aims to analyze the factors affecting student productivity by using data analytics and visualization techniques. The system uses Power BI to transform raw student data into meaningful dashboards.
- Data Collection:
  - Gather student productivity dataset containing study hours, screen time, sleep hours, stress levels, grades, and lifestyle factors.
  - Collect information related to digital activities such as phone usage, gaming hours, and social media usage.
- Data Preprocessing:
  - Clean the dataset by removing missing or inconsistent values and the column that are not required.
  - Convert data into correct formats and organize fields such as study hours, sleep hours, and productivity scores.
- Data Analysis & Visualization:
  - Use Microsoft Power BI to analyze relationships between different variables.
  - Identify patterns between study habits, lifestyle factors, and productivity levels.
- Deployment:
  - Design an interactive dashboard with multiple pages including.
    - Overview Dashboard.
    - Study Behaviour Analysis.
    - Digital Distraction Analysis
    - Lifestyle & Health Impact Analysis

  - Result: The dashboard provides clear insights into how study habits, digital distractions, and lifestyle factors influence student productivity and academic performance.

SYSTEM APPROACH:

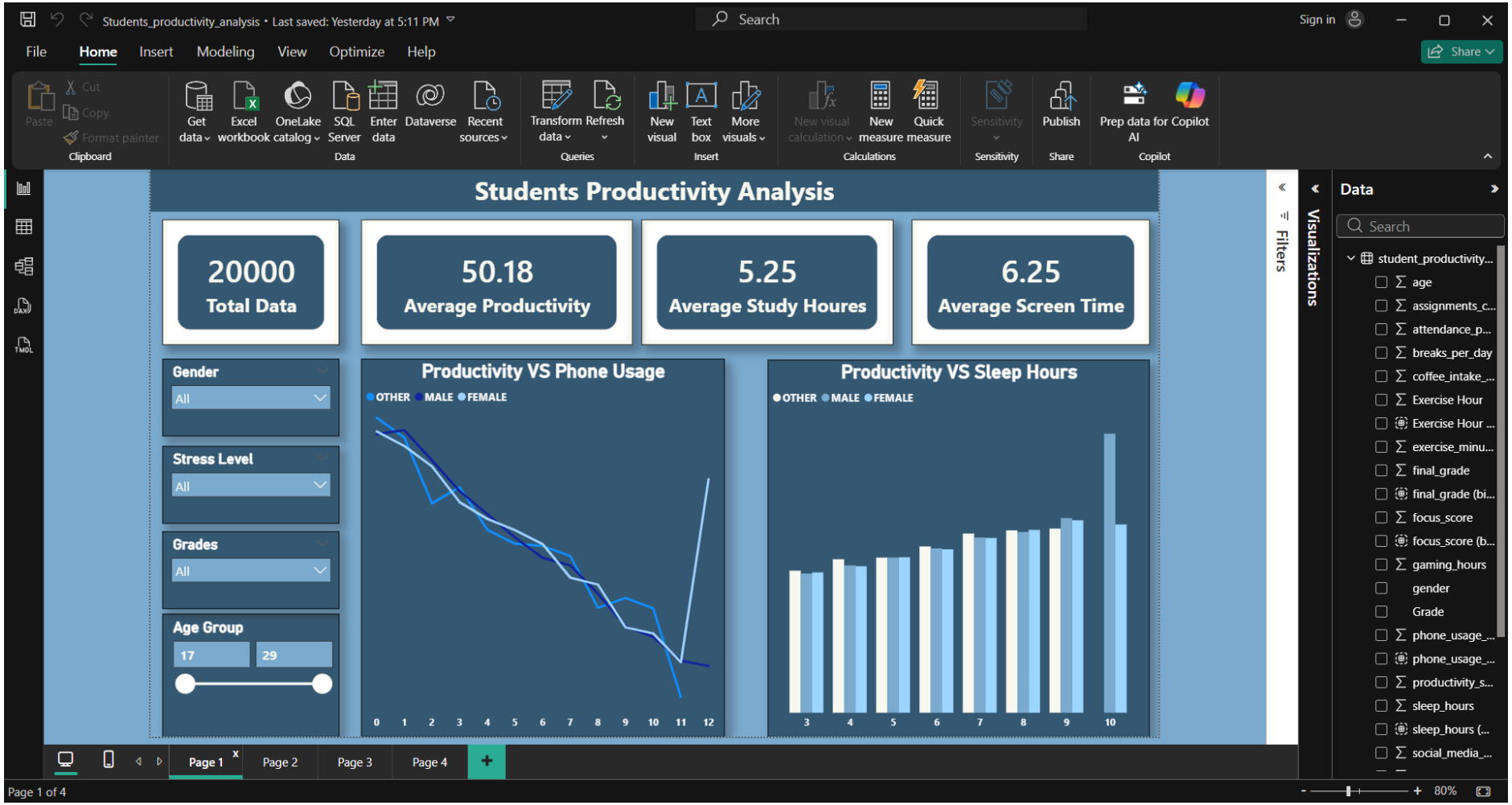
- **Data collecting:**
  - Collected Data from online source Kaggle.
- **Data cleaning:**
  - Missing values are removed and data types are corrected.
- **Data Analysis:**
  - The dataset is analyzed in Microsoft Power BI to find patterns.
- **Data Visualization:**
  - Charts such as line charts, bar charts, and pie charts are created.
- **Dashboard Creation:**
  - An interactive dashboard with filters (gender, age group, stress level, grades) is built.

ALGORITHM & DEPLOYMENT:

- In this project, data analytics techniques are used to analyze student productivity and understand how study habits, digital distractions, and lifestyle factors influence academic performance.
- Algorithm Selection:
  - Power BI analytics and visualization techniques are used to analyze relationships between study hours, screen time, sleep hours, and productivity.
- Data Input:
  - The dataset includes study hours, phone usage, screen time, sleep hours, stress level, grades, and productivity scores.
- Analysis Process:
  - The dataset is cleaned and prepared for analysis.
  - Charts such as line charts, column charts, and scatter plots are used to identify patterns.
- Deployment:
  - An interactive Power BI dashboard is created to visualize the results.
  - Filters such as gender, age group, and stress level allow users to explore the data dynamically.
- Result:
  - The dashboard provides insights into how different factors affect student productivity and academic performance.

RESULT:

The Power BI dashboard successfully visualizes student productivity data and identifies relationships between study habits, digital distractions, and lifestyle factors.



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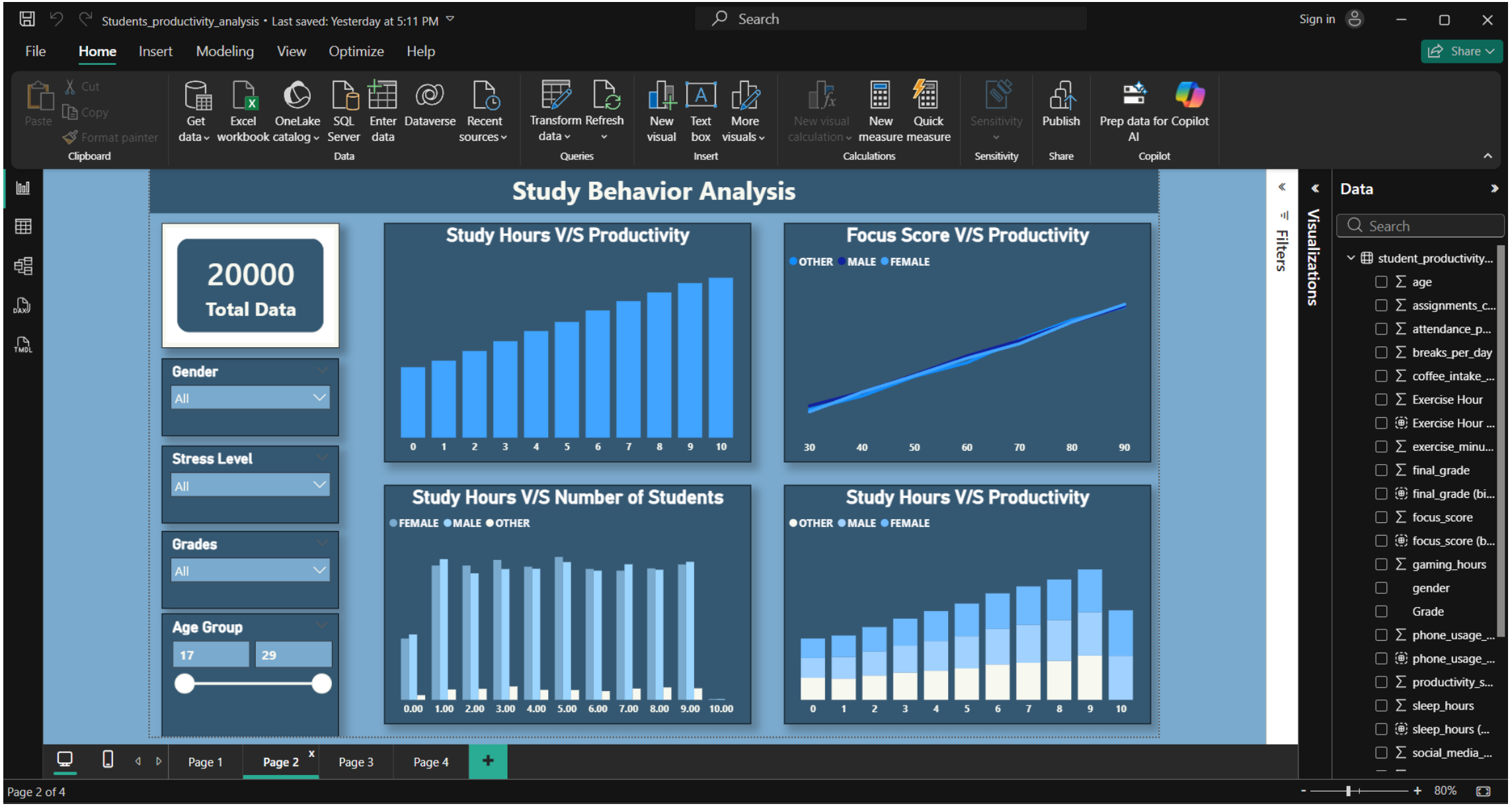

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CONCLUSION:

- This project analyzes student productivity using Microsoft Power BI to understand how study habits, digital distractions, and lifestyle factors affect academic performance.
- The analysis shows that factors such as study hours, phone usage, sleep patterns, and stress levels have a significant impact on student productivity.
- Interactive dashboards and visualizations help users easily explore the relationships between these variables and identify important patterns.
- Overall, the Power BI dashboard provides useful insights that can help students and educators better understand behaviors that improve academic productivity.

FUTURE SCOPE:

- The project can be improved by adding more student-related data such as attendance, learning methods, and academic history to provide deeper insights into productivity.
- Advanced analytics and machine learning techniques can be integrated to predict student performance and identify productivity patterns more accurately.
- The dashboard can also be expanded to analyze larger datasets from different schools or institutions to support better educational decision-making.
- Additional features such as real-time data updates and automated reports can further enhance the usability of the Power BI dashboard.

REFERENCES:

- GitHub Repository Link :

  Link

- Datasheet From Kaggle:

  Link

- My LinkedIn Profile Link:

  Link

Thank You